

Safe control for a coupled two-axis robot — technical memo

Recommendation. Take the frozen deterministic baseline (fingerprint 7d857df507c389c9) to hardware qualification, and do not deploy the tested adaptive load correction.

- **In simulation:** the baseline tracks the A–C scenarios to 0.3–0.7° RMS with no faults and slows the loaded D/E requests rather than saturating blindly.
- **Infeasible requests:** it restricts or rejects those it can see; an unknown overload is attempted first, then caught and suspended.

Labels: *Observed* = the brief's five summaries only; *Calculated* = the supplied model; *Simulated* = our simulator, not hardware; *Proposed* = not yet done.

uv run python run_all.py regenerates all evidence in a locked environment (verified on macOS arm64 and Linux x86-64). The full answers are in [task1–task5](#).

1. Understanding the failure

What the motor permits (Calculated).

- **Torque ceiling:** $K_t I_{\max} = 0.448 \text{ N}\cdot\text{m}$ at 3.2 A and 0.336 N·m derated.
- **Run A has headroom:** its 2.1 A peak is 0.294 N·m, 66% of the ceiling. At 45°, gravity plus the friction and disturbance allowances need only 0.175 N·m. So A's 2.8° RMS error points to response, not current capacity (voltage and trajectory feasibility are not excluded).
- **Yaw coupling ($\pm 75^\circ$):** $0.008 \dot{q}_y + 0.0008 \ddot{q}_y$ peaks at 0.136 N·m at 1.5 Hz (B) and 0.247 N·m at 2.2 Hz (C). Including the 0.05 N·m disturbance and friction allowances, B needs 50% of nominal capacity. C needs 75% of nominal but 100.3% of derated: holdable at 3.2 A, marginal at 2.4 A.
- **Payload:** the 35 mm centre-of-mass shift adds 0.343·m N·m. The mass is not given, so D/E feasibility cannot be settled from the summaries.

Leading explanation (hypothesis). The controller neither anticipates nor rejects the disturbance torque within its current limit.

- **B→C:** modelled coupling rises 1.82×; observed RMS rises 1.86× and peak error 1.80×.
- **D:** its +4.6° signed mean is 26% of the mean-square error, consistent with an unmodelled gravity load.
- **E:** 25% less torque gives 38% more error, assuming E keeps D's payload, which the brief does not state.

What the summaries cannot prove: the original controller (E's cycling does not prove windup), payload mass and direction, trajectory timing, what "clipped" counts, voltage use and thermal history. Our reconstructed legacy controller matches RMS ordering only to about 25%: a consistency check, not identification.

The experiment that would most change this view: hold roll near zero and sweep yaw in frequency within a qualified envelope. Log synchronized encoders, current command and measurement, active limit, timestamps and bus voltage. Fit the roll torque residual against yaw velocity and acceleration, and validate at a held-out frequency.

2. Baseline controller

Element	Choice and reason
Rates	500 Hz position loop, the brief's maximum. Phase loss is dominated by delay (1 ms command delay, 0.6–1.8 ms CAN, 1.2 ms current lag; 7 ms design, 11 ms burst corner), so a faster loop gains little. 1 kHz drive-side supervisor.
Feedback	PI × lead, $C(s) = 0.7553(1 + 2.8/s)(1 + s/7)/(1 + s/112)$, Tustin-discretized; 4.46 Hz crossover.
Feed-forward	Nominal inertia, gravity and 80% friction on the <i>governed</i> reference. Yaw coupling from a causal Kalman estimate of the yaw encoder, predicted 4 ms ahead; no future yaw plan.
Estimates	14-bit encoder, 0.022° per count. One count differentiated at 500 Hz is 0.19 rad/s, ten times the friction speed scale, so velocity is filtered and friction compensation uses reference velocity.
Saturation	Feed-forward-priority clamp, conditional-integration anti-windup, target clamped to the latest 3.2/2.4 A limit after the delay queue.
Governor	Admits motion within $0.8 \cdot K_t I_{lim} - 0.05$ N·m (nominal load model).
Faults	Communication loss → drive-local damping; re-arm after 50 ms of fresh aligned commands. Error > 12° for 40 ms → latched fault, host catch, request suspended until replanned. 3 latches in 30 s → lockout. Overspeed and thermal trips.

When to reshape, derate or reject (governor rules):

- **Reshape** (slow the path clock) when dynamic torque exceeds the budget but every pose is holdable.
- **Derate** the budget as soon as the current limit drops.
- **Restrict or reject** poses whose static holding demand exceeds capacity; slowing cannot fix a static overload.
- **Request yaw reduction**, or suspend with a coordinated stop, when coupling exceeds roll authority.
- **Never** shift a stationary reference to absorb a disturbance.

Why it should stay well behaved.

- **Margins (Calculated):** 49.4° phase margin and 13.5 dB gain margin at 7 ms; 32.1° and 6.1 dB at the 11 ms corner with $J -30\%$ and $K_t +15\%$; 37.8 ms pure-delay margin.
- **Stress tests (Simulated):** 60 ms outages in either or both directions, burst storms, 5% message loss, the combined corner, and 20 V with $R +25\%$ are event-free or re-arm automatically. The 20 V fast sweep is reshaped to 58% progress.
- **Sampled plants (20 per scenario):** A/B/C have no events but meet the tracking threshold in 13, 20 and 15 of 20; D/E have events in 5 and 6 of 20.
- **Quantization:** about 31 mA of current jitter in Run B (7.6 mA at 24 bits), with no instability. Hunting at hold is about 0.2° of friction stick-slip.
- **Not proven:** stability under deep saturation with an unknown load, the main residual risk given negative gravity stiffness.

3. Learning: evaluated and rejected

Candidate. A bounded two-parameter load residual, $\theta_s \sin q + \theta_c \cos q$.

- **What it observes:** delayed measured roll/yaw, measured current, the active limit and health timestamps, during healthy, settled, yaw-stationary 250 ms dwells.
- **Timing:** it updates at ≤ 50 Hz on a non-blocking worker (2–8 ms latency) and discards results older than 100 ms.

- **What it changes:** feed-forward only, ≤ 0.20 N·m. The governor, limits and faults keep the nominal model.
- **When it is ignored:** on stale, invalid, faulted, saturated, mismatched, under-covered or expired data.

Registered comparison (Simulated). The protocol, untuned held-out loads (5 loads $\times$ 2 limits $\times$ 5 seeds), metric (RMS over the first 0.5 s of each test dwell) and gate are cited in a commit made before the candidate was run; the plan file itself was first committed together with the results. The gate requires $\geq 10\%$ median paired improvement over the best retuned deterministic comparator, $\geq 80\%$ availability, and no safety or completion regression.

50 held-out pairs	Median primary	vs int1	Completed
int1: retuned comparator (= frozen baseline)	2.130°	—	40/50
int2: faster integrator, fails margin criteria	1.157°	+42.6%	50/50
Adaptive candidate	2.088°	0.0%	40/50
Known-load feed-forward oracle (not deployable)	0.499°	+76.9%	50/50

- **The 0.0% is exact:** the candidate is int1 plus the learned term on the same seed. In 30 of 50 pairs the correction never changed a scored command: never usable (13), or first usable at 18.3–21.6 s (17), at the end of the test where bumpless transfer cancels a change at constant reference. These pairs are bit-identical to int1.
- **When it was on in time, it helped:** 20 pairs improved, 8 of them by 42–79%, and none got worse (mean +9.5%, not registered).
- **Gate:** fails on benefit and on availability (74%). No safety criterion fails in the held-out runs or in 96 registered challenge runs (yaw B/C, feedback loss, derating). But the correction was active in only 23/48 candidate challenge runs, and never in the 9 s-onset yaw B/C sets, so most challenge runs test the baseline. Mid-run payload change was not tested.
- **Incomplete runs:** the 10 in each variant reach every waypoint but overshoot the $\pm 65^\circ$ range by 5.65–6.89° (5° allowed).

What would change the choice: an estimator that is reliably usable during calibration (the oracle shows the load law is worth about 77%), passing a newly registered held-out comparison including payload change.

4. Evidence

Simulator: 10 kHz plant integration; current lag, command delay, CAN latency with bursts and loss; 14-bit encoders; multirate timing; current and voltage limits (R $1.8 \Omega \pm 25\%$, L $0.45 \text{ mH} \pm 20\%$, $K_e = K_t$, bus to 20 V). Uncertainty covers J $\pm 30\%$, $K_t \pm 15\%$, friction, payload, coupling and latency.

Scenario (5-seed median)	Legacy RMS	Final governed RMS	Original-request RMS	Progress	At limit	Tracked
A shaped $\pm 45^\circ$	3.50°	0.68°	0.74°	100%	0.0%	5/5
B yaw 1.5 Hz	3.57°	0.27°	0.27°	100%	0.0%	5/5
C yaw 2.2 Hz	7.64°	0.40°	0.40°	97%	0.0%	5/5
D payload $\pm 80^\circ$	10.23°	4.10°	76.7°	83%	1.9%	0/5
E D at 2.4 A	12.02°	4.25°	78.5°	51%	5.5%	0/5

- **Legacy** is a reconstruction; both controllers are simulated.

- **D/E:** the governed error is small only because the requests are slowed. The original-request error shows the sacrificed timing, and D/E count as not tracked (tracked = $\geq 95\%$ progress, $\leq 2^\circ$ RMS, no fault; a project threshold).

Right answer is not to track:

- **INF-P:** 1.2 kg at +35 mm, derated, needs 0.41 N·m of static torque against 0.336 N·m available. The nominal-model baseline cannot know this: it attempts the move, sags to about -58° , trips, and is held near -49° after the catch, suspended. This is a reported limitation. A truth-informed governor rejects such a load outright.
- **INF-R:** roll hold under 2.8 Hz yaw. The governor reduces yaw to $0.36\times$ and holds roll within $\pm 1.1^\circ$.
- **Loaded M2:** trips and stays suspended rather than cycling.
- **Fallback:** local damping does not hold against gravity; a loaded 100 ms outage moves the axis up to 42° .

Reproducibility. A fresh `git` archive snapshot installed from `uv.lock` reproduced every result value, `run_id` and figure exactly on the Mac. On Linux x86-64, all outcomes were identical and values agreed to $3e-11$ (NumPy rounds random draws differently across CPUs) ([R4](#)). 262 unit tests pass.

5. Testing the explanation

Registered prediction (Calculated; the registration and the results were first committed together, so git gives no ordering evidence). In Run B, weaken the true K_t and K_ϕ by 10% with the controller unchanged. Predicted: coupling peak 0.1356 N·m (0.969 A), extra ideal current +0.108 A (± 0.15 A), and an uncanceled coupling residual of +0.0136 N·m (± 0.01).

Result (Simulated: 3 seeds, feed-forward on/off pairs, 2–8 s). A windowing and unit bug in our first analysis was found and corrected retrospectively ([R2](#)).

- **Consistency checks only:** the coupling and ideal-current matches re-check the model's own arithmetic.
- **Not tested:** the extra current is not separable in the total current, and its tolerance includes 0.
- **Failed:** the residual increase was -0.0007 N·m, because the causal yaw estimate's transient error (about 0.03 N·m) swamps it. With exact feed-forward (a diagnostic) the predicted +0.0123 N·m appears.
- **Supported:** feed-forward stays valuable, and removing it costs $3.5\text{--}3.9^\circ$.
- **Not uniform:** the weaker motor raised error in only 2 of 3 pairs.

Revised explanation: at this operating point, residual error is dominated by estimator transients; motor strength is second-order.

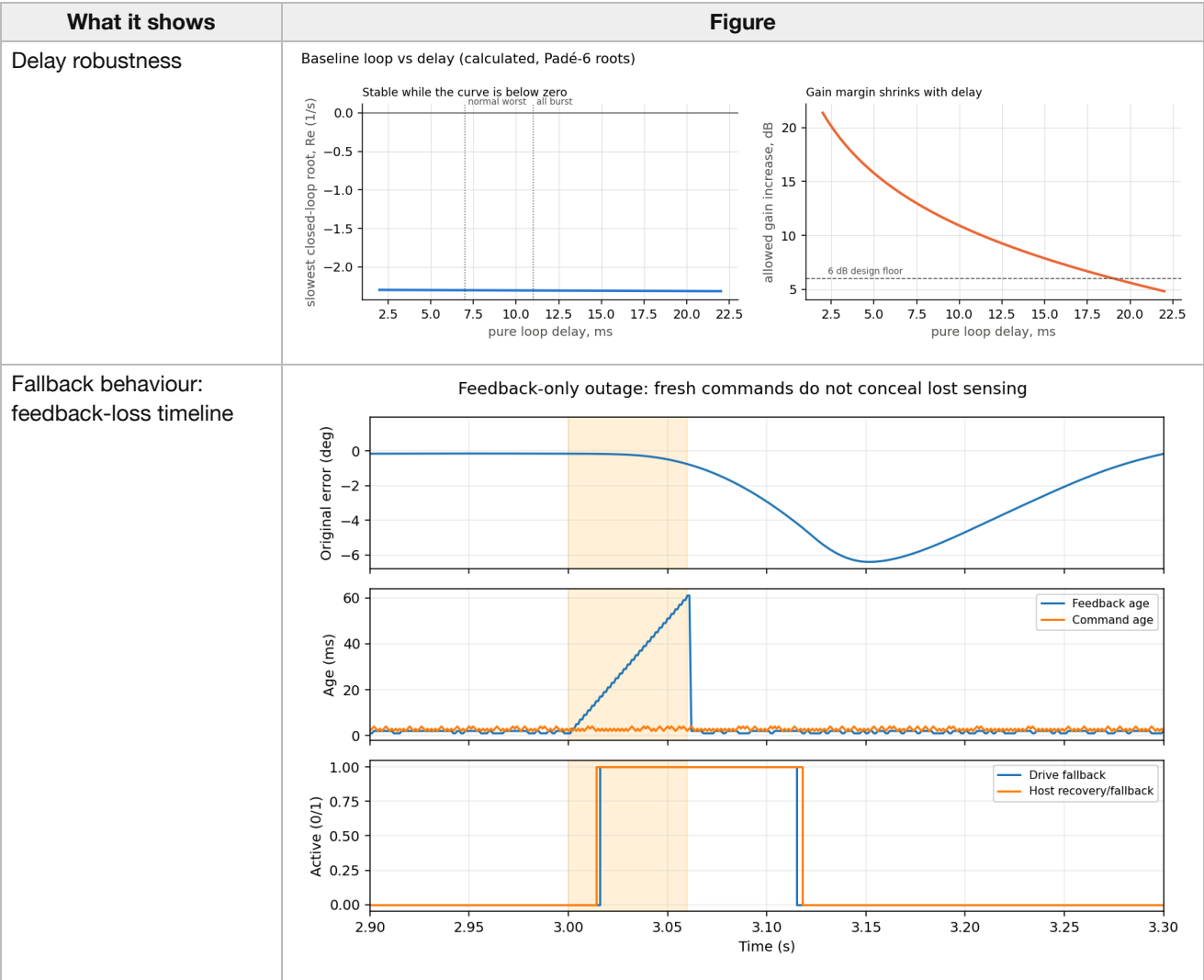
Smallest justified design change: none to the frozen controller. If hardware calibration finds that the effective K_t differs from nominal, rescale the feed-forward current by $K_{t,nom}/K_{t,meas}$. That is one parameter, and it is proposed, not tested.

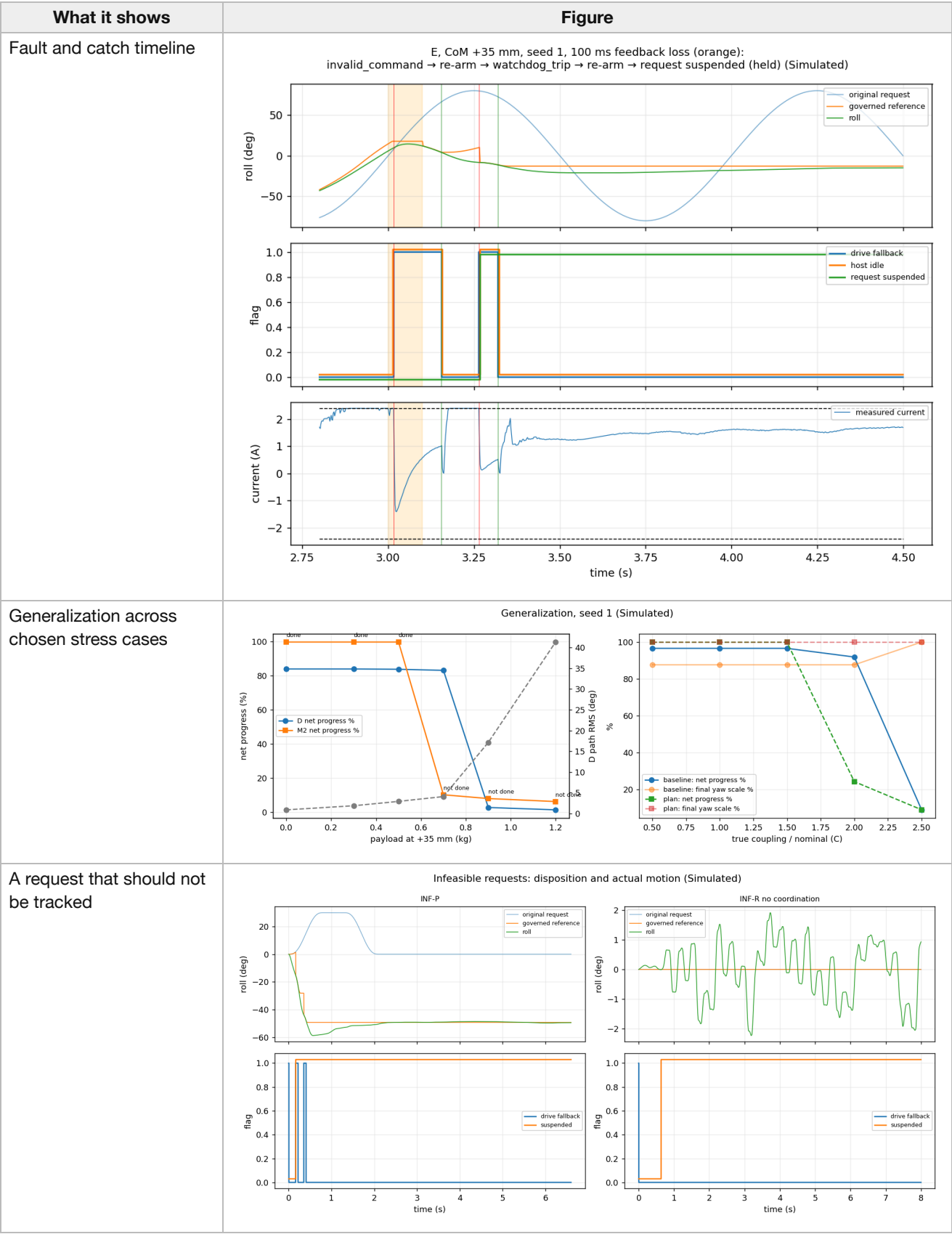
Hardware confirmation: torque-versus-current fixture identification, then the roll-held yaw sweep ([qualification plan](#)).

Risks carried to hardware: the drive voltage convention (V_{bus} vs $V_{bus}/\sqrt{3}$), the loaded fallback excursion, admission of unknown-load moves that later trip, and assumed thermal thresholds.

Appendix A — Plots (outside the page limit)

What it shows	Figure
Saturation and anti-windup	30° requested in 0.1 s: saturation with and without governor / anti-windup (simulated) The top plot shows roll in degrees over 2 seconds. The bottom plot shows current A over the same period. Four scenarios are compared: A (governor ON, real supervisor), B (governor OFF, anti-windup ON, no watchdog), C (governor OFF, anti-windup OFF, no watchdog), and D (governor OFF, anti-windup ON, real supervisor). roll ° current A time s A. governor ON, real supervisor (the baseline) B. governor OFF, anti-windup ON, no watchdog C. governor OFF, anti-windup OFF, no watchdog D. governor OFF, anti-windup ON, real supervisor
Phase lag: loop frequency response and delay margins	Roll tracking response, yaw still, d(t) off (amplitude 10° down to 2° at 6 Hz) The top plot shows the magnitude ratio q / q_req in dB on a linear scale. The bottom plot shows the phase in degrees on a linear scale. The x-axis for both is frequency in Hz on a logarithmic scale from 0.1 to 20 Hz. Four scenarios are compared: baseline (calculated linear), bench (simulated), baseline (simulated), and legacy (simulated). q / q_req (dB) phase (deg) frequency (Hz) - baseline, Calculated (linear) - bench, Simulated - baseline, Simulated - legacy, Simulated





What it shows

Tracking, original vs governed reference

Figure

